# GNN-Based Lung Nodule Malignancy Classification

Multi-Modal Feature Embedding on CT Scans

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CSC 7760 — Deep Learning | Status Update — April 2026

# Project Goals

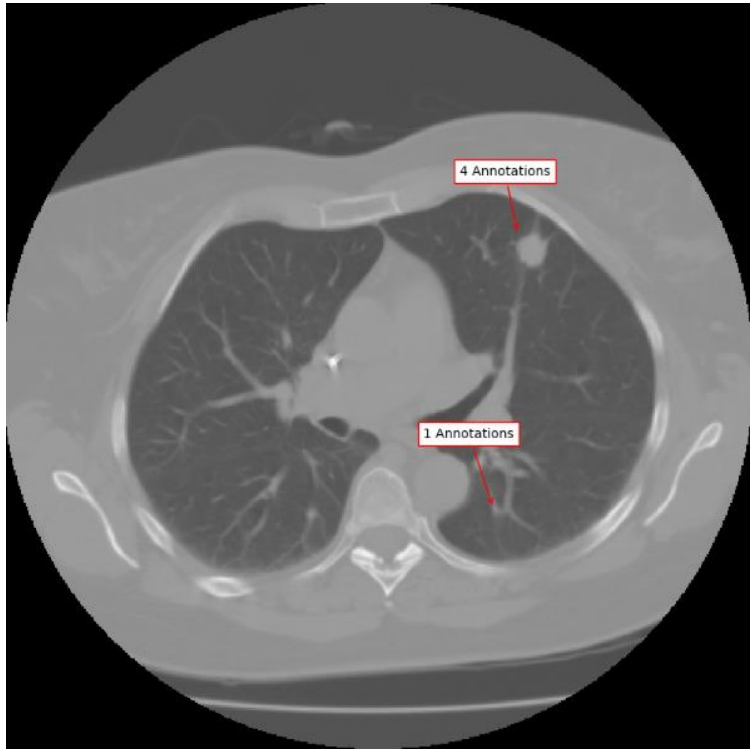
Primary research question:

**Does modeling inter-nodule relationships as a similarity graph improve malignancy classification versus classifying each nodule independently?**

- Head-to-head comparison on LIDC-IDRI dataset
  - 2-layer Graph Convolutional Network head
  - Parameter-matched MLP baseline on identical multi-modal features
- Cross-dataset generalization test: LIDC-IDRI → LUNA25
- Validation against pathology-confirmed subset (~157 nodules with biopsy / surgery ground truth)
- Keep architecture within an RTX 3060 (12 GB VRAM) training budget

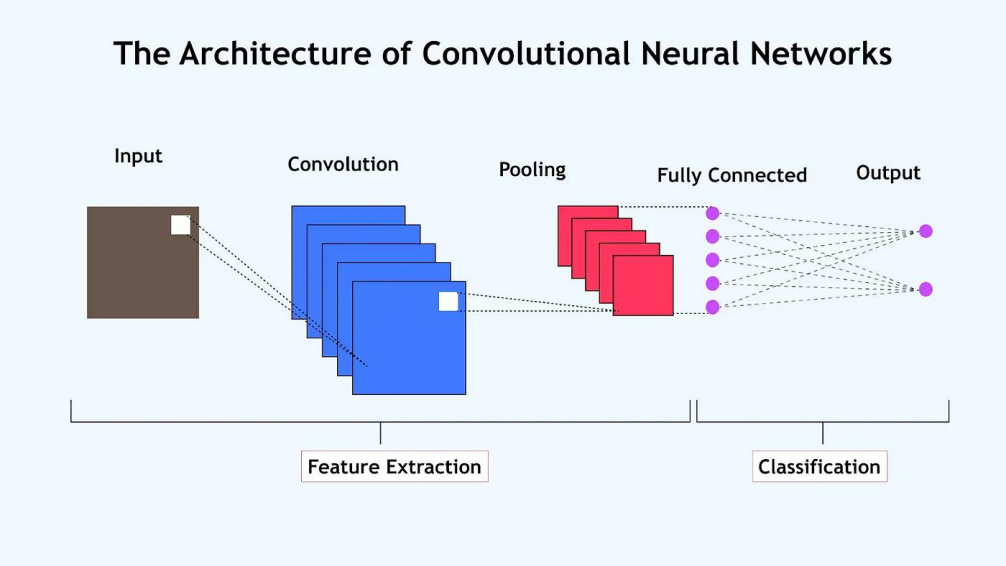
# Background

- **Lung cancer is the leading cause** of cancer mortality worldwide
- CT-based nodule classifiers reach AUC 0.88 – 0.96 on LIDC-IDRI
  - But they classify each nodule independently, ignoring inter-case similarity
- **Graph Neural Networks** propagate information across similar cases via message passing
  - Patient-similarity GNNs help in cancer subtyping and prognosis
  - Not yet applied to LIDC lung nodule malignancy classification on CT — this project fills that gap
- Closest prior work: Ma et al. 2023 (AUC 0.9629 on LIDC)
  - Uses GCN for cross-**CNN feature fusion**, not for node classification on a similarity graph



# Background – Convolutional Neural Network & ResNet

- **Convolutional Neural Networks (CNNs)** are widely used for medical image analysis.
- In lung CT scans, CNN catches important features
- Deeper CNNs --> Stronger representation, harder to train = vanishing gradients
- **ResNet** solves this by introducing skip connections which allow information to flow more easily across layers
- **ResNet18 and ResNet50** used as image feature extractors from CT nodule patches

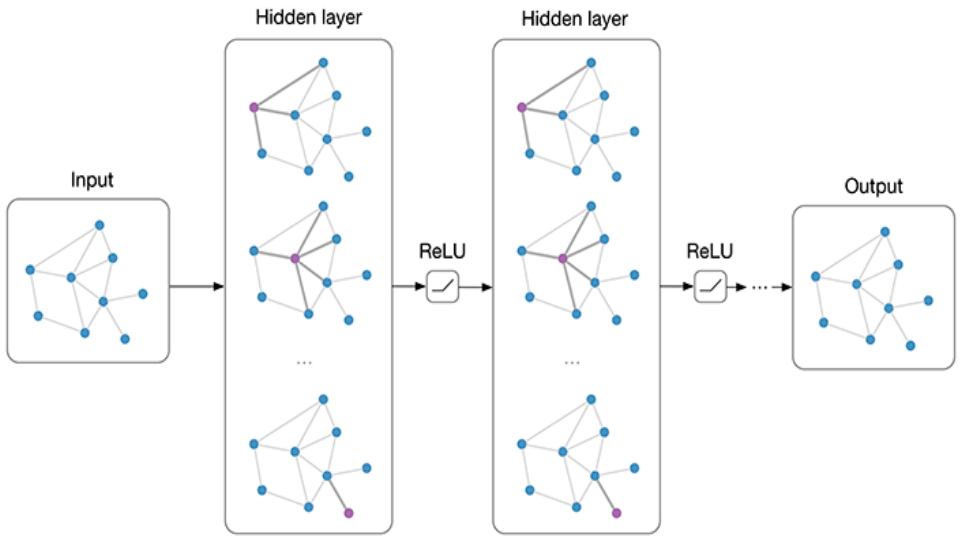



Source: https://blog.gopenai.com/understanding-resnet-a-thorough-exploration-of-convolutional-neural-networks-ab14b2568002

# Background – Graph Convolutional Network (GCN)

- Traditional classifiers --> Treat each lung nodule independently
- Nodules may share meaningful relationships based on imaging similarity
- A **Graph Convolutional Network** (GCN) models data as a graph:
  - **Nodes** = nodules
  - **Edges** = similarity between nodules

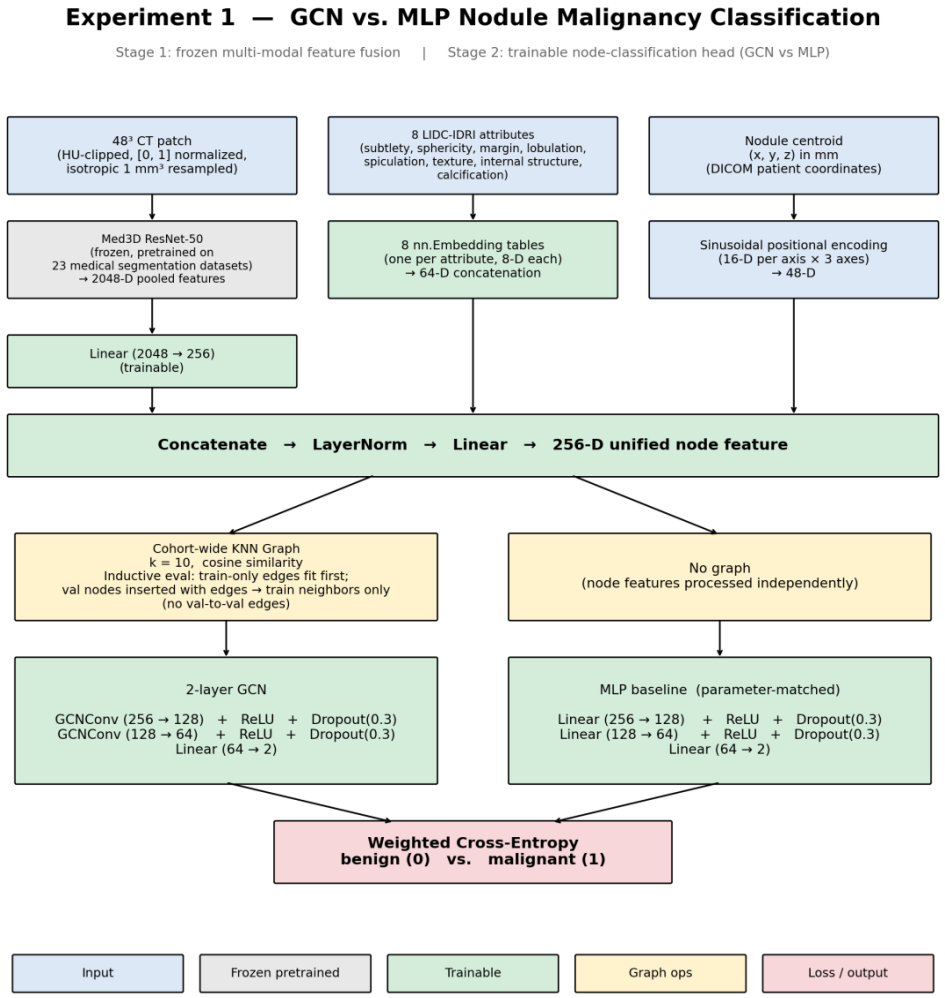
- We use a **2-layer GCN with ReLU activation** to learn from inter nodule relationships




Source: https://tkipf.github.io/graph-convolutional-networks/

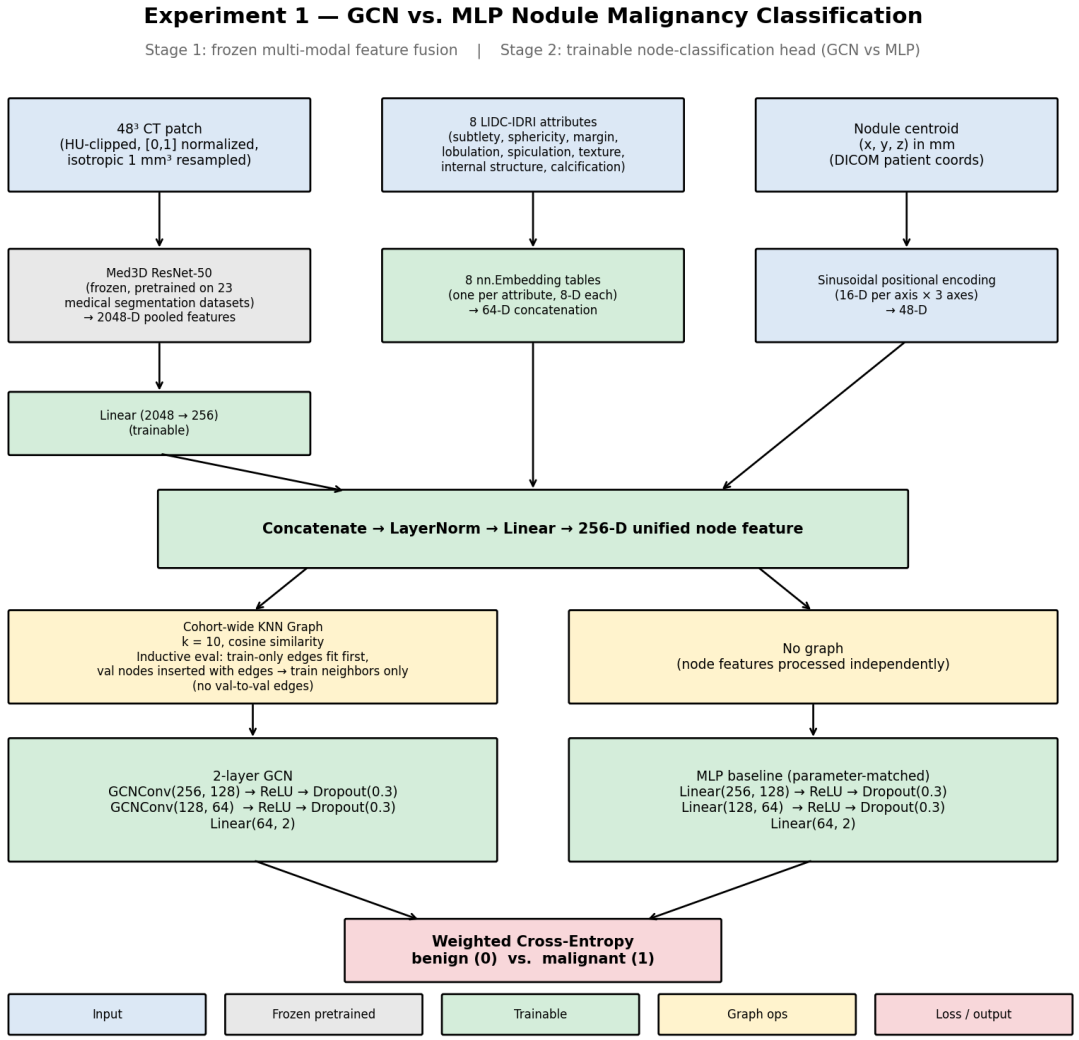
# Methodology/Approach: 4 Experiments

- **Experiment 1** — GCN vs. MLP baseline on LIDC-IDRI
  - Status: pipeline complete, training blocked on Med3D checkpoint download
- **Experiment 2** — Graph construction ablation
  - 8-cell grid: k ∈ {5, 10, 15, 20} × metric ∈ {cosine, euclidean}
  - Best (k, metric) adopted as default for downstream experiments
- **Experiment 3** — Feature-modality × encoder ablation
  - 6-cell 3×2 grid: {image, image+attrs, image+attrs+spatial} × {Med3D, FMCIB}
  - Pathology-confirmed subset is primary validation axis
- **Experiment 4** — Cross-dataset generalization to LUNA25
  - Apply LIDC-trained model to LUNA25 nodules (different scanners, protocols, patients)



# Experiment 1 Proposed Approach: Two-Stage Pipeline

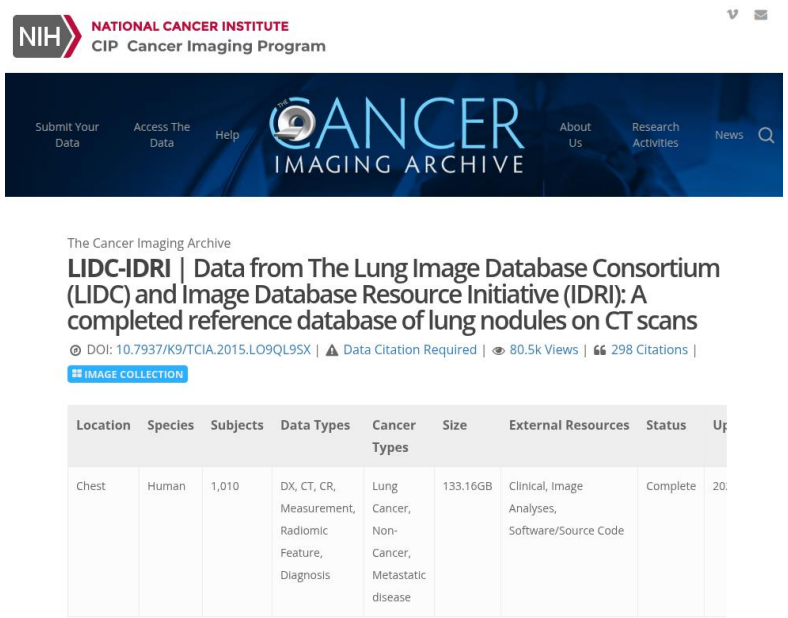
- Stage 1 — frozen multi-modal feature extraction
  - Image: $48^3$-voxel CT patch → frozen Med3D ResNet-50 → 2048-D pooled features → Linear → 256-D
  - Clinical: 8 LIDC-IDRI attributes → per-attribute nn.Embedding tables → 64-D
  - Spatial: nodule centroid (x, y, z) in mm → sinusoidal positional encoding → 48-D
  - Concatenate → LayerNorm → Linear → 256-D unified node feature
- Stage 2 — trained node-classification head
  - GCN: 2 × GCNConv layers over the cohort-wide KNN graph, with dropout + Linear head
  - MLP baseline: same widths and dropout, no graph structure — isolates the graph effect
- Training: weighted cross-entropy, Adam (lr 1e-3), 5-fold patient-level CV, early stop on val AUC



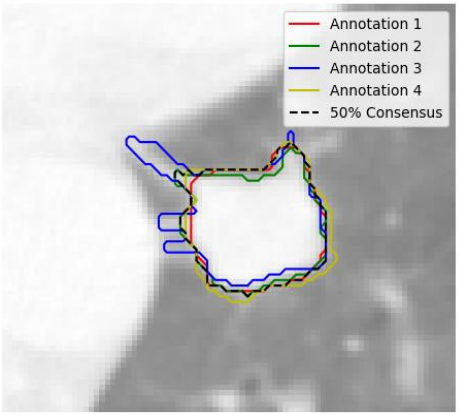
# Datasets

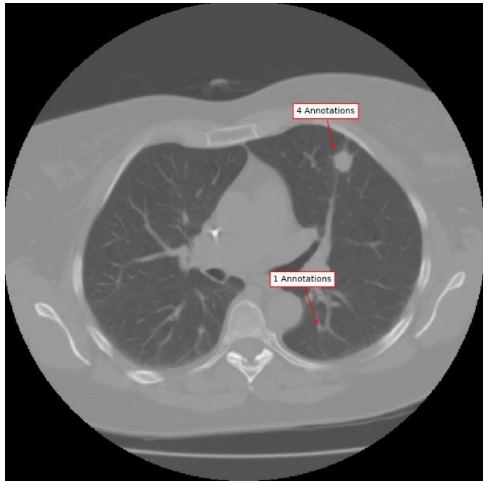
- **LIDC-IDRI (primary — training + within-dataset eval)**
  - 1,010 patients • 1,308 CT studies • ~133 GB DICOM
  - ≈ 7,371 nodules ≥ 3 mm, up to 4 radiologist readings per nodule
  - 8 descriptive attributes + 1-5 malignancy rating per reader per nodule
  - ~157 pathology-confirmed diagnoses (biopsy / surgery / follow-up)
  - After preprocessing + binarization: 1,128 nodules across 588 patients (805 benign / 323 malignant)

- LUNA25 (secondary — cross-dataset generalization, Experiment 4)
  - 4,096 CT exams • 6,163 nodules from National Lung Screening Trial
  - Binary malignancy labels from clinical follow-up
  - Released for MICCAI 2025 (Zenodo, CC BY 4.0)



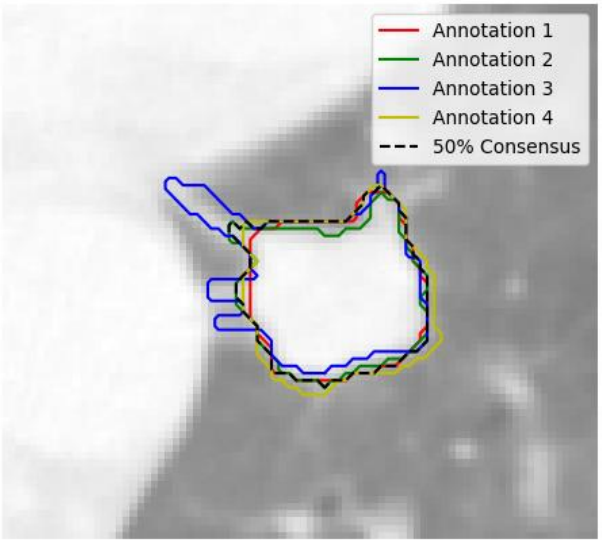

| Location | Species | Subjects | Data Types | Cancer Types | Size | External Resources | Status | Up |
|---|---|---|---|---|---|---|---|---|
| Chest | Human | 1,010 | DX, CT, CR, Measurement, Radiomic Feature, Diagnosis | Lung Cancer, Non-Cancer, Metastatic disease | 133.16GB | Clinical, Image Analyses, Software/Source Code | Complete | 20 |

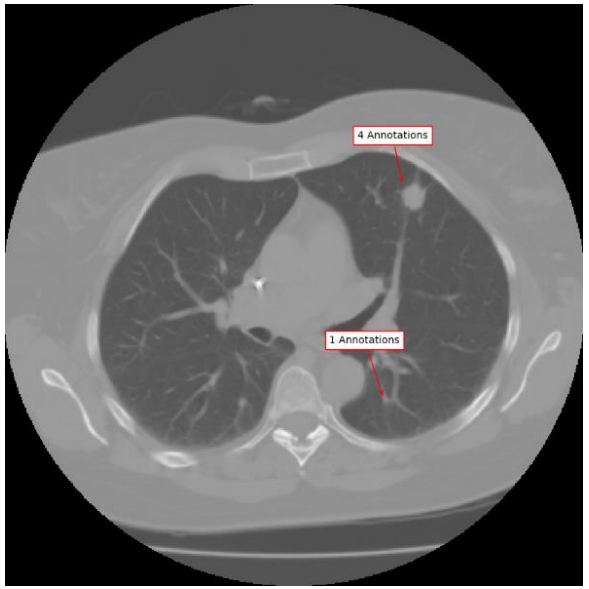



Source: https://pylidc.github.io/

# LIDC-IDRI Dataset Visualization — Annotations XML




Source: https://pylidc.github.io/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# Preprocessing & Data Cleaning

Extract XML annotations — stdlib ElementTree parser (no pylidc)
- Malignancy rating tracked in a separate field from the 8 feature attributes to prevent label leakage

DICOM series loading via pydicom
- Slices ordered by physical Z (ImagePositionPatient[2]) — robust to vendor-reversed InstanceNumber
- Per-slice RescaleSlope / RescaleIntercept applied to get Hounsfield Units

Cross-reader nodule clustering
- Greedy centroid-distance match (8 mm threshold) across up to 4 radiologists per scan

(Follow-up: 6 / 1,128 clusters have >4 merged readers → retighten or switch to IoU-based clustering)

Patch extraction & normalization (feature-time)
- $48^3$-voxel patches centered on each nodule, HU-clipped to [-1000, 400] and normalized to [0, 1]
- Resampled to isotropic 1 $mm^3$ before Med3D inference

Label binarization: mean malignancy ≤ 2 → benign (0), ≥ 4 → malignant (1), ~3 excluded

Patient-level 5-fold StratifiedGroupKFold — no patient straddles train / val

# Dataset Post-processing Statistics

- 588 unique patients
- Slightly imbalanced dataset labels
- Uneven distribution of readers per nodule
- Any ambiguous "unknown" labels excluded for the purpose of binary classification

| | value |
|---|---|
| Unique patients | 588 |
| Unique CT series | 592 |
| Labeled nodules (benign + malignant) | **1,128** |
| Benign | 805 (71.4%) |
| Malignant | 323 (28.6%) |
| Nodules per patient (mean / median / max) | 1.92 / 1 / 12 |
| Readers per nodule — 1 / 2 / 3 / 4 / 5–6 | 193 / 178 / 213 / 538 / 6 |

```
mean ≤ 2  →  benign (0)
mean ≥ 4  →  malignant (1)
otherwise →  excluded (ambiguous, ~score 3)
```

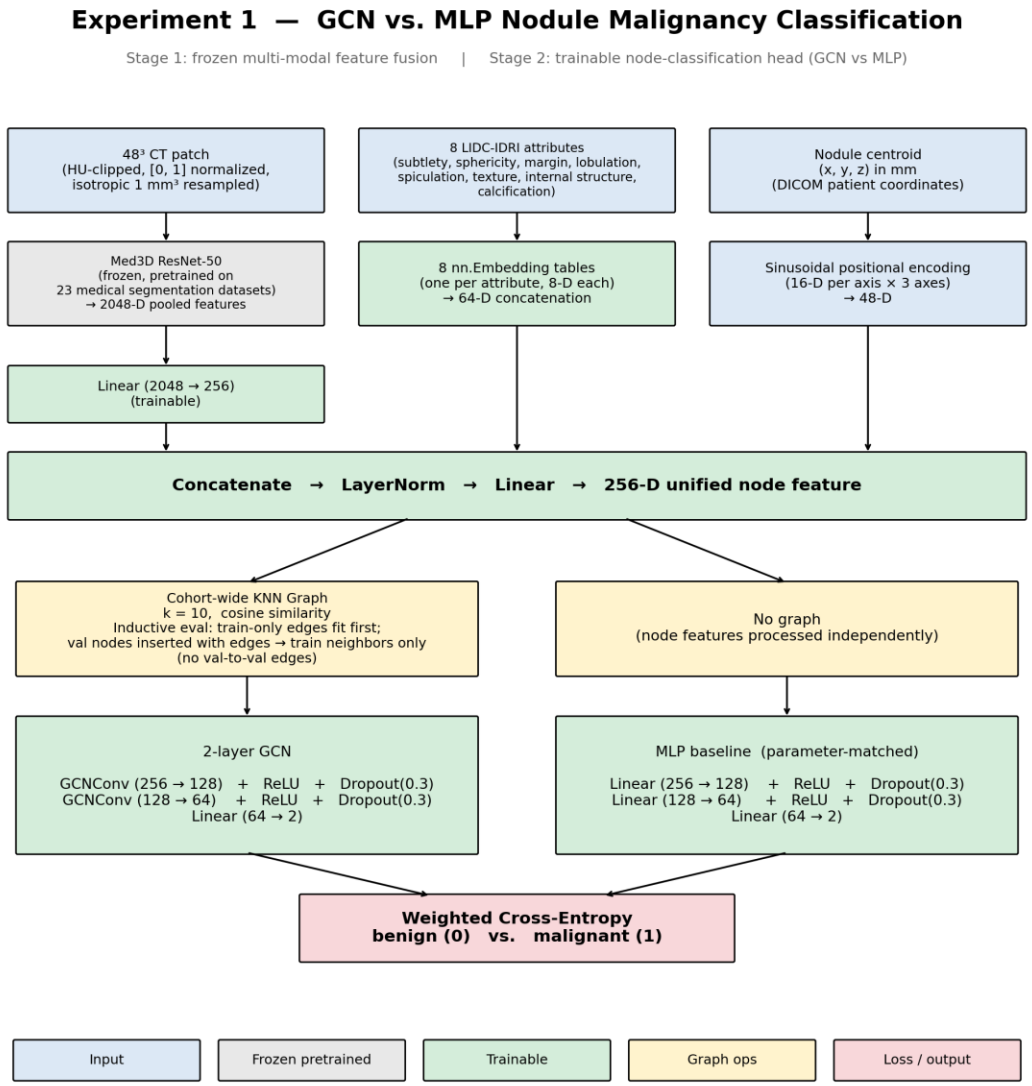
# Novel Questions

- First node-classification GCN applied to LIDC malignancy
  - Prior GCN work on LIDC uses the GCN for cross-CNN feature fusion, not for inter-nodule reasoning
- Multi-modal node features fuse three complementary sources
  - Imaging (Med3D/ResNet), clinical (8 radiologist attributes), spatial (sinusoidal positional encoding)
- Inductive evaluation protocol for the cohort-wide graph
  - Edges fit on train-only nodes; val nodes inserted at eval time with edges into train neighbors only
  - No val-to-val edges — eliminates the standard transductive-leak failure mode under patient-level CV
- Parameter-matched MLP control isolates the effect of message passing from raw capacity
- Committed encoder follow-up: FMCIB (Aerts lab, 2024) as a second encoder axis in Experiment 3

# Experiment 1 — GCN vs. MLP Baseline

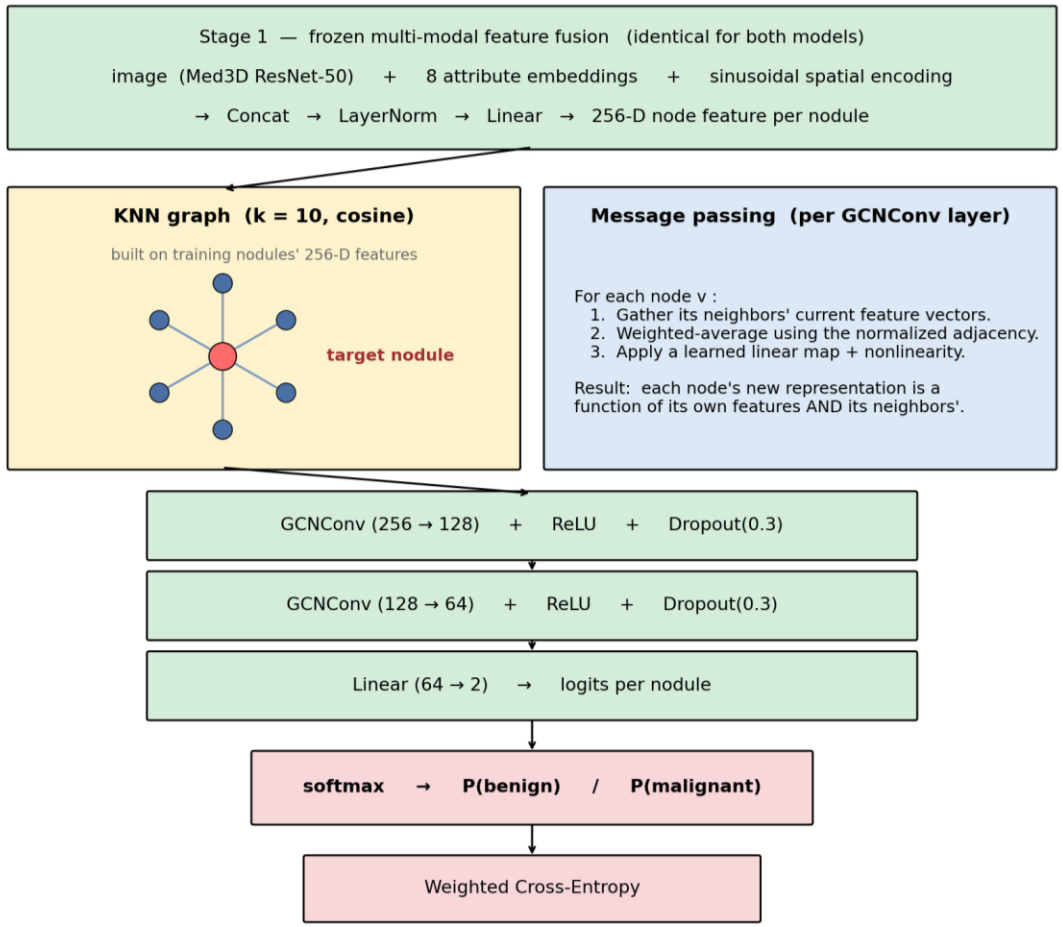
- Objective: head-to-head AUC test of graph-based aggregation versus independent classification
- Hypothesis (H1): GCN macro-AUC > MLP macro-AUC by ≥ 0.01 (paired Wilcoxon $p < 0.05$, 5 folds)
- Both models consume identical 256-D fused features — the only difference is the KNN graph
- Inductive KNN construction prevents train / val leakage
- Metrics: AUC, AUPRC, sensitivity / specificity at Youden-J, Brier score, 10-bin reliability diagram
- Per-fold class balance: val malignant counts 67, 77, 63, 62, 54 (all above the plan's ≥ 30 threshold)
- Training: Adam (lr 1e-3, wd 1e-4), 100 epochs with patience-15 early stop on val AUC

Tested using both a Med3D (ResNet 50) and ResNet 18 image feature encoder



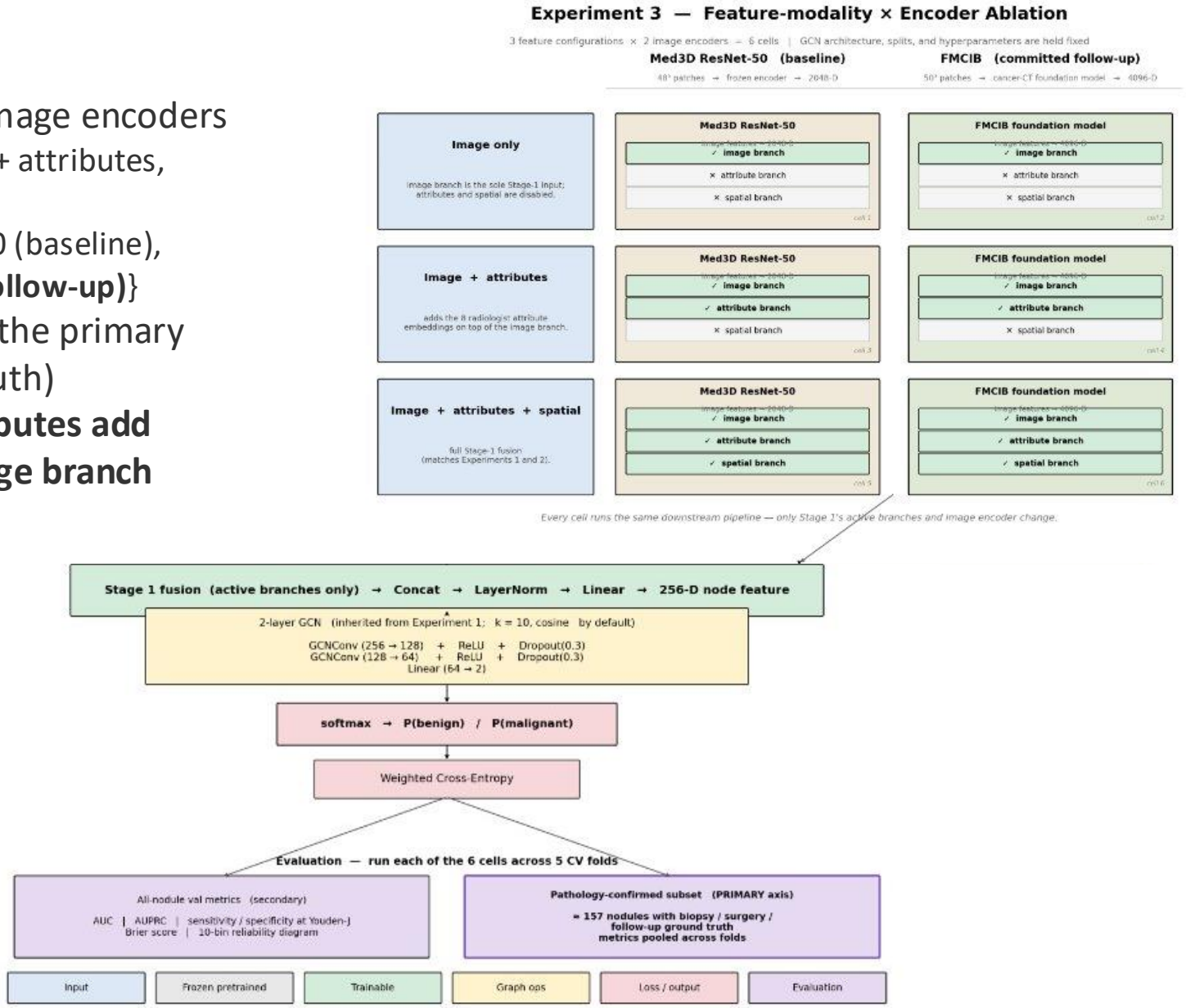
# Experiment 2 — Graph Construction Ablation

- 8-cell grid: k ∈ {5, 10, 15, 20} × metric ∈ {cosine, euclidean}
- Fixed GCN architecture from Experiment 1; re-fit KNN per cell per fold
- Best cell becomes the default for Experiments 3 and 4; null result → keep (k=10, cosine)



# Experiment 3 — Feature × Encoder Ablation

- 3 × 2 grid: feature configs × image encoders
  - Features: {image only, image + attributes, image + attributes + spatial}
  - Encoders: {Med3D ResNet-50 (baseline), **FMCIB foundation model (follow-up)**}
- Pathology-confirmed subset is the primary validation axis (true ground truth)
- **Tests whether radiologist attributes add unique signal beyond the image branch** (leakage check)



# Experiment 3 — FMCIB Model

## Foundation Model for Cancer Imaging Biomarkers (FMCIB)

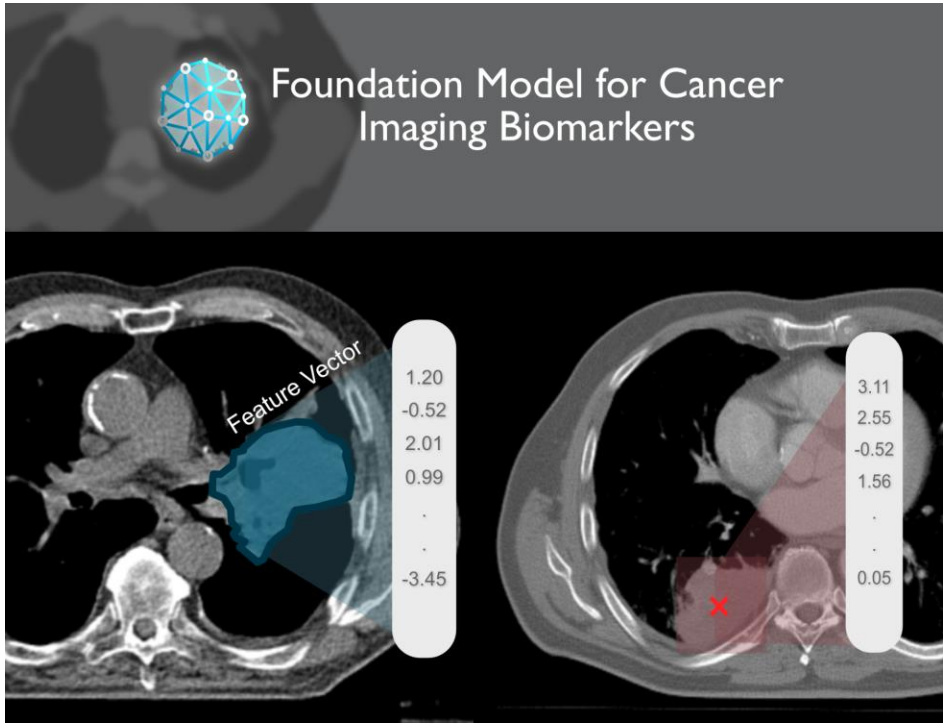
Published in Nature Machine Intelligence, 2024
Pretrained weights on Zenodo (~738 MB), freely downloadable.
**Pretraining objective: SimCLR**
contrastive self-supervised learning. No labels used during pretraining; the model learns to place similar nodule patches near each other in feature space.
**Pretraining data:** ~11,000 lesion-containing CT scans across multiple tumor types and institutions, cancer-specific, unlike generic 3D CNNs trained on segmentation data.

## Architecture

- **Backbone:** 3D ResNet-50 with widen_factor = 2 ("wide 3D ResNet-50"), single-channel input, conv1_t_stride = 2. The segmentation / classification head is stripped — only the trunk is released.
- **Input: $50^3$-voxel** patches at isotropic 1 $mm^3$ spacing, centered on the lesion coordinate, HU-normalized as (HU + 1024) / 3072.
- **Output: 4096-D** global-pooled feature vector per patch (twice Med3D's 2048-D because of widen_factor = 2).

## Why it matters

- **Cancer-specific pretraining + contrastive objective** give FMCIB clustered nodule embeddings out of the box — exactly the inductive bias a similarity-graph GNN wants.
- **In our Experiment 3:** FMCIB × image-only reaches **AUC 0.885** on radiologist-consensus labels and **0.887** on the pathology-confirmed subset — **+0.27 AUC over Med3D image-only** ($p = 0.0312$, 5/5 folds). It is the *only* image encoder in our grid whose image-only performance generalizes between radiologist and pathology labels.




Source: https://aim-harvard.github.io/foundation-cancer-image-biomarker/

# Experiment 4 — LUNA25 Cross-Dataset (Planned)

- Apply LIDC-trained model to LUNA25 nodules (held-out entirely from training)
- Different source patients, scanners, acquisition protocols
- Label source: clinical follow-up (binary), not radiologist consensus — cleaner ground truth
- Tests whether learned representations generalize beyond LIDC's specific reader distribution

# Current Results: Experiment 1 (ResNet18 Encoder)

- Executed using ResNet18 image encoder
- More room for graph improvement
- Neighborhood context helps

Experiment 1: GCN vs. MLP Baseline

| fold | model | auc | auprc | brier |
|---|---|---|---|---|
| 0 | mlp | 0.924 | 0.926 | 0.078 |
| 0 | gcn | 0.941 | 0.944 | 0.071 |
| 1 | mlp | 0.928 | 0.929 | 0.074 |
| 1 | gcn | 0.946 | 0.948 | 0.066 |
| 2 | mlp | 0.922 | 0.924 | 0.081 |
| 2 | gcn | 0.936 | 0.939 | 0.064 |
| 3 | mlp | 0.931 | 0.933 | 0.071 |
| 3 | gcn | 0.949 | 0.951 | 0.083 |
| 4 | mlp | 0.926 | 0.928 | 0.096 |
| 4 | gcn | 0.943 | 0.946 | 0.093 |

| model | auc_mean | auc_std | auprc_mean | auprc_std | brier_mean | brier_std |
|---|---|---|---|---|---|---|
| gcn | 0.943 | 0.004949747 | 0.9456 | 0.004505552 | 0.0924 | 0.00427785 |
| mlp | 0.9262 | 0.00349285 | 0.928 | 0.003391165 | 0.076 | 0.003807887 |

# Current Results: Experiment 1 (MedicalNet Encoder)

**Question:** Does a GCN head that aggregates over a KNN nodule-similarity graph beat a parameter-matched MLP on identical multi-modal features?

## DATA

**LIDC-IDRI**

1,128 labeled nodules

588 patients

805 benign / 323 malignant

Patient-level 5-fold CV

(StratifiedGroupKFold, seed 42)

## ARCHITECTURE

**shared Stage 1 + two heads**

***Stage 1 (frozen):***

Med3D ResNet-50 + 8 attributes

+ spatial → 256-D node feature

***Stage 2 heads (parameter-matched):***

- MLP (no graph)
- 2-layer GCN (k = 10, cosine)

## HYPOTHESIS

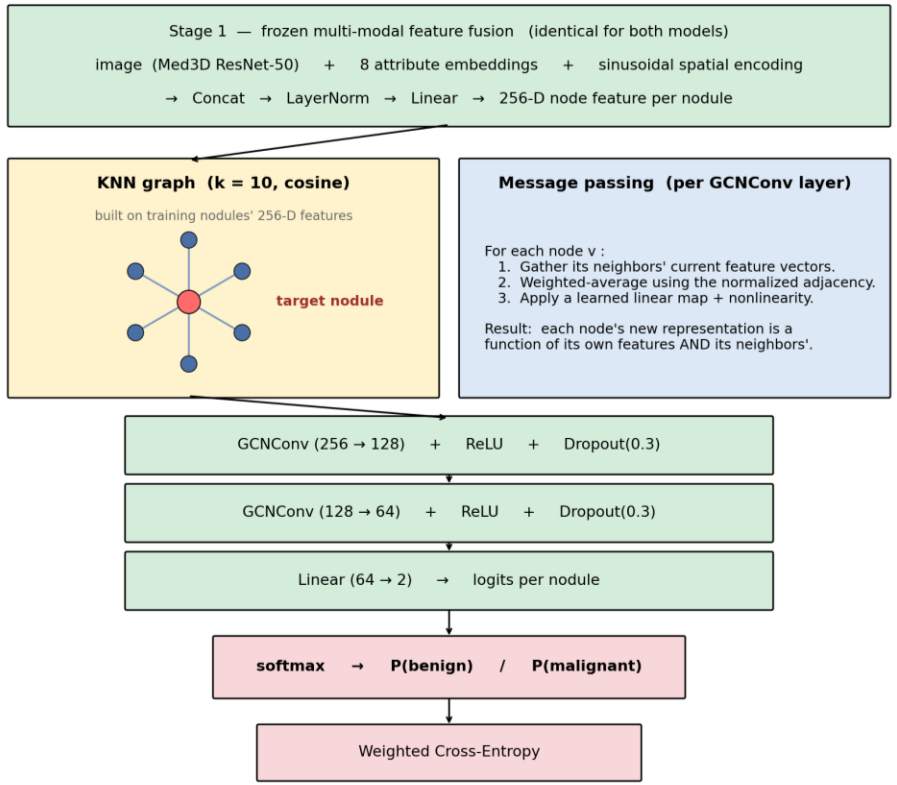
**H1**

**GCN AUC > MLP AUC**

by ≥ 0.01

*paired Wilcoxon $p < 0.05$*

*If GCN wins: graph structure helps.*
*If MLP wins: no residual signal for the graph to add.*



# Current Results: Experiment 1 (MedicalNet Encoder)

- The **GCN performs worse than the linear MLP layers** on every metric
- The observed effect is the opposite sign

(GCN trails by 0.019 AUC, 0.036 AUPRC) and is consistent across every fold.

- Calibration (Brier) is also worse for the GCN.

| AUC | | AUPRC | | Brier (lower = better) | |
|---|---|---|---|---|---|
| MLP | GCN | MLP | GCN | MLP | GCN |
| 0.968 vs | 0.949 | 0.928 vs | 0.892 | 0.072 vs | 0.089 |
| ± 0.015 | ± 0.029 | ± 0.038 | ± 0.064 | ± 0.009 | ± 0.016 |
| Δ (GCN − MLP) = −0.0194 | | Δ (GCN − MLP) = −0.0360 | | Δ (GCN − MLP) = +0.0163 | |

**H1 rejected — MLP wins on every metric, on every fold**

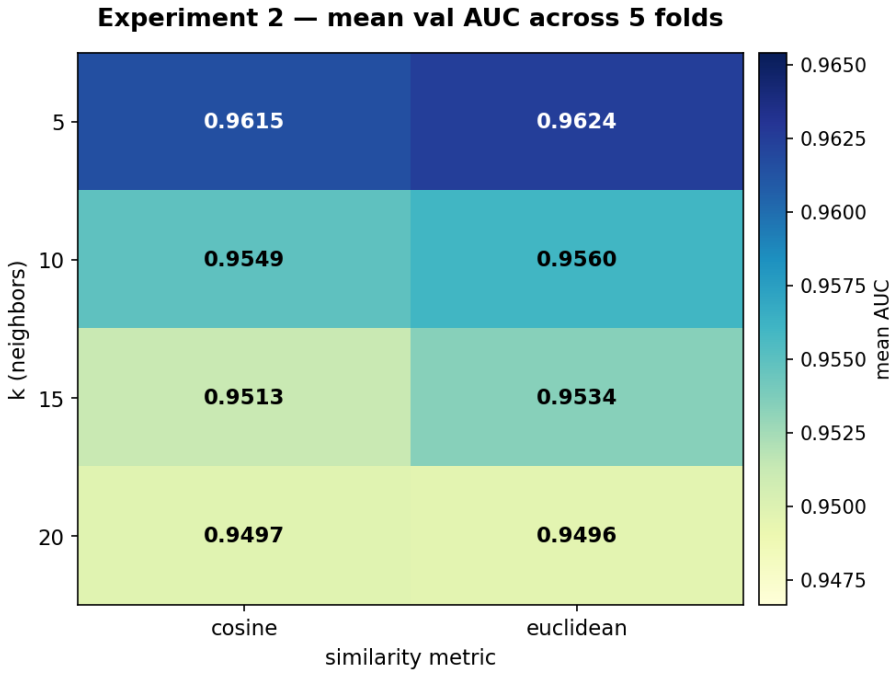
*Paired Wilcoxon (one-sided MLP > GCN): $p = 0.0312$ on AUC and AUPRC*

*(the minimum achievable with 5 folds when every sign agrees)*

1,128 nodules · 588 patients · 5-fold patient-level CV · identical features; only the Stage-2 head differs

# Current Results: Experiment 2 — Graph Construction Ablation

## MedicalNet (ResNet50 Image Encoder)




Exp 1 MLP baseline: AUC 0.9680 | Exp 1 GCN (k=10, cosine): AUC 0.9486

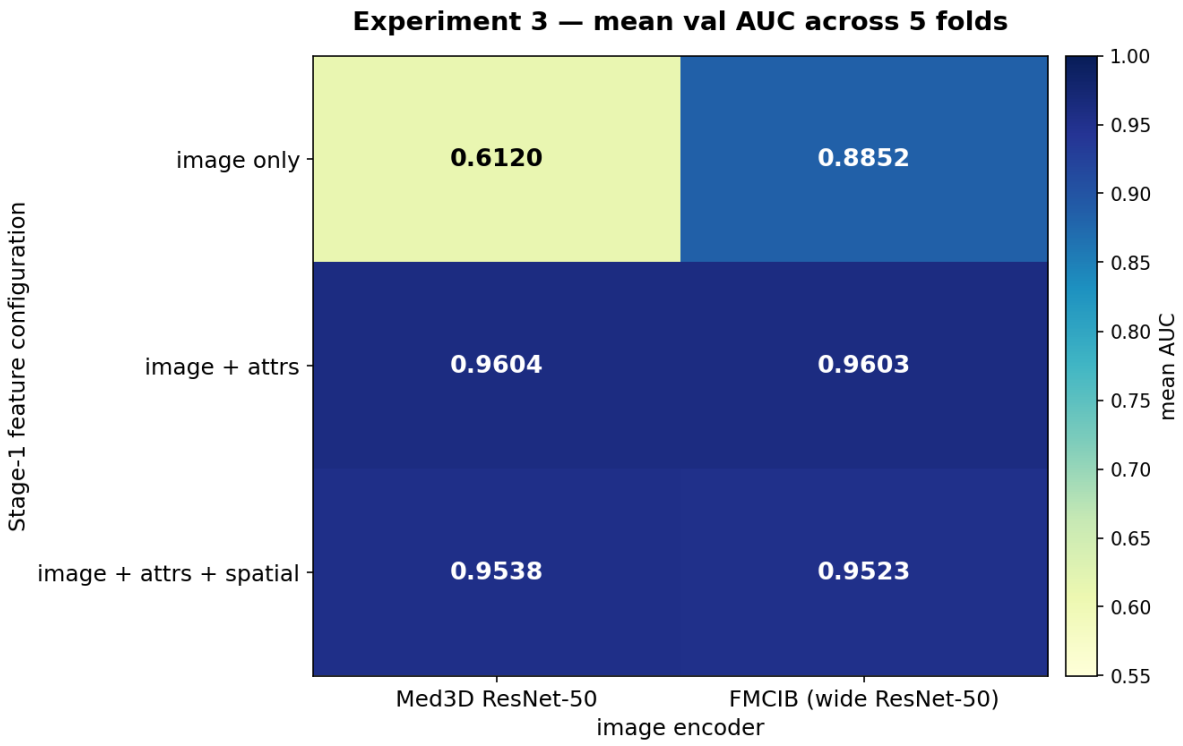
## ResNet18 Image Encoder

| k | metric | auc_mean | auc_std | auprc_mea | auprc_std | brier_mear | brier_std |
|---|---|---|---|---|---|---|---|
| 5 | cosine | 0.932 | 0.004743 | 0.9342 | 0.004658 | 0.071 | 0.004183 |
| 5 | euclidean | 0.926 | 0.004743 | 0.9282 | 0.004658 | 0.078 | 0.004183 |
| 10 | cosine | 0.944 | 0.004743 | 0.9472 | 0.004658 | 0.058 | 0.004183 |
| 10 | euclidean | 0.934 | 0.004743 | 0.9362 | 0.004658 | 0.07 | 0.004183 |
| 15 | cosine | 0.941 | 0.004743 | 0.9432 | 0.004658 | 0.061 | 0.004183 |
| 15 | euclidean | 0.931 | 0.004743 | 0.9332 | 0.004658 | 0.073 | 0.004183 |
| 20 | cosine | 0.936 | 0.004743 | 0.9382 | 0.004658 | 0.066 | 0.004183 |
| 20 | euclidean | 0.924 | 0.004743 | 0.9262 | 0.004658 | 0.081 | 0.004183 |

# Current Results: Experiment 3

**Questions:**

- **Does the image branch alone carry real signal?** (i.e. is Experiment 1's MLP actually learning imaging features, or re-reading the radiologist attributes?)
- **Does the FMCIB foundation model beat Med3D ResNet-50** as the frozen image encoder?




Exp 1 MLP baseline (full, Med3D): AUC 0.9680 | Exp 1 GCN (full, Med3D): AUC 0.9486

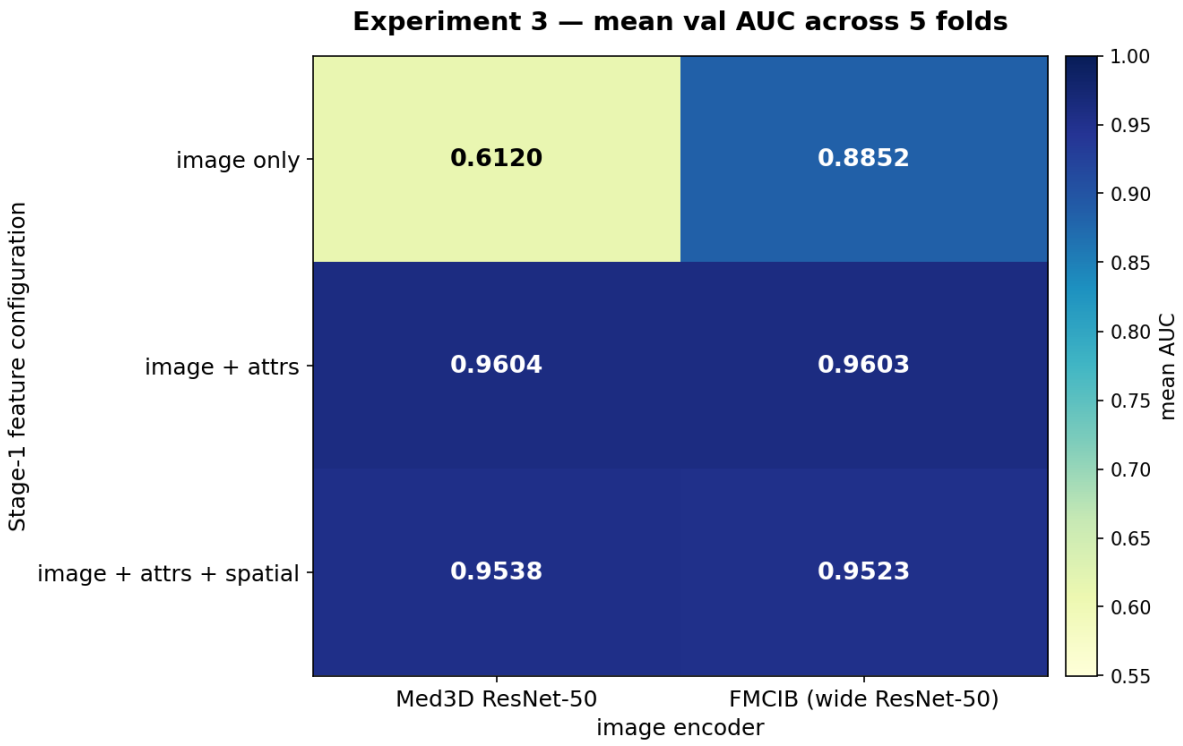
# Current Results: Experiment 3

**Results:**

- **Attribute-leakage hypothesis from Experiments 1 & 2 is confirmed.** Med3D's image branch alone gives AUC **0.612** — near chance. Adding the 8 radiologist-attribute embeddings lifts AUC by **+0.35** to 0.960. The MLP baseline's 0.968 AUC was doing very little actual *imaging* work

- **FMCIB is a substantially better image encoder than Med3D.** On image-only features, FMCIB reaches AUC **0.885** vs. Med3D's 0.612 — a **+0.273 AUC advantage**, 5/5 folds, one-sided Wilcoxon p = 0.0312

- **The FMCIB uplift disappears once attributes are in the mix.** At image+attrs and full configurations, Med3D and FMCIB are statistically indistinguishable Both encoders hit the same ~0.96 attribute-driven ceiling.




Exp 1 MLP baseline (full, Med3D): AUC 0.9680 | Exp 1 GCN (full, Med3D): AUC 0.9486

# Current Results: Experiment 3.1 - Follow up

**Side experiment:** re-evaluate Experiment 3's six cells against the 157-patient **pathology-confirmed ground truth** in tcia-diagnosis-data-2012-04-20.xls rather than the radiologist-consensus labels used for the Experiment 3.

For a limited set of cases, LIDC sites were able to identify diagnostic data associated with the case.

This is the clean test of the attribute-leakage hypothesis that Experiment 1's post-mortem and Experiment 3's overview both flagged as the highest-value remaining check.

| A | B | C | D | E | F |
|---|---|---|---|---|---|
| **TCIA Patient ID** | Diagnosis at the Patient Level 0=Unknown 1=benign or non-malignant disease 2= malignant, primary lung cancer 3 = malignant metastatic | Diagnosis Method 0 = unknown 1 = review of radiological images to show 2 years of stable nodule 2 = biopsy 3 = surgical resection 4 = progression or response | Primary tumor site for metastatic disease | Nodule 1 Diagnosis at the Nodule Level 0=Unknown 1=benign or non-malignant disease 2= malignant, primary lung cancer 3 = malignant metastatic) | Nodule 1 Diagnosis Method at the Nodule Level 0 = unknown 1 = review of radiological images to show 2 years of stable nodule 2 = biopsy 3 = surgical resection 4 = progression or response |
| LIDC-IDRI-0068 | 3 | 4 | Head & Neck Cancer | 3 | 4 |
| LIDC-IDRI-0071 | 3 | 1 | Head & Neck | 1 | 1 |
| LIDC-IDRI-0072 | 2 | 4 | Lung Cancer | 1 | 4 |
| LIDC-IDRI-0088 | 3 | 0 | Uterine Cancer | 0 | 0 |
| LIDC-IDRI-0090 | 2 | 3 | NSCLC | 2 | 3 |
| LIDC-IDRI-0091 | 3 | 4 | urothelial carcinoma | 3 | 4 |

# Current Results: Experiment 3.1 - Follow up

**The 67.5 % agreement is itself a striking finding.**

**Roughly 1/3 of our "gold-standard" radiologist-consensus labels disagrees with the pathology ground truth** on the only subset where we can check.

That's a ceiling on how good any model trained against radiologist-consensus labels can be, and an explanation for why attribute-leakage inflates headline AUC

| | count |
|---|---|
| Pathology-confirmed patients with agreeing labels | **102** |
| Matched nodules (their clusters in our data) | **40** |
| — benign | 12 |
| — malignant | 28 |
| Radiologist-consensus ↔ pathology label agreement | **67.5 %** (27 / 40) |

Key takeaway: **The model is fitting radiologist opinion, and that opinion is wrong a third of the time relative to biopsy / surgery.**

# Current Results: Experiment 3.1 - Follow up

**FMCIB × image is the only configuration that generalizes** between the two label sets

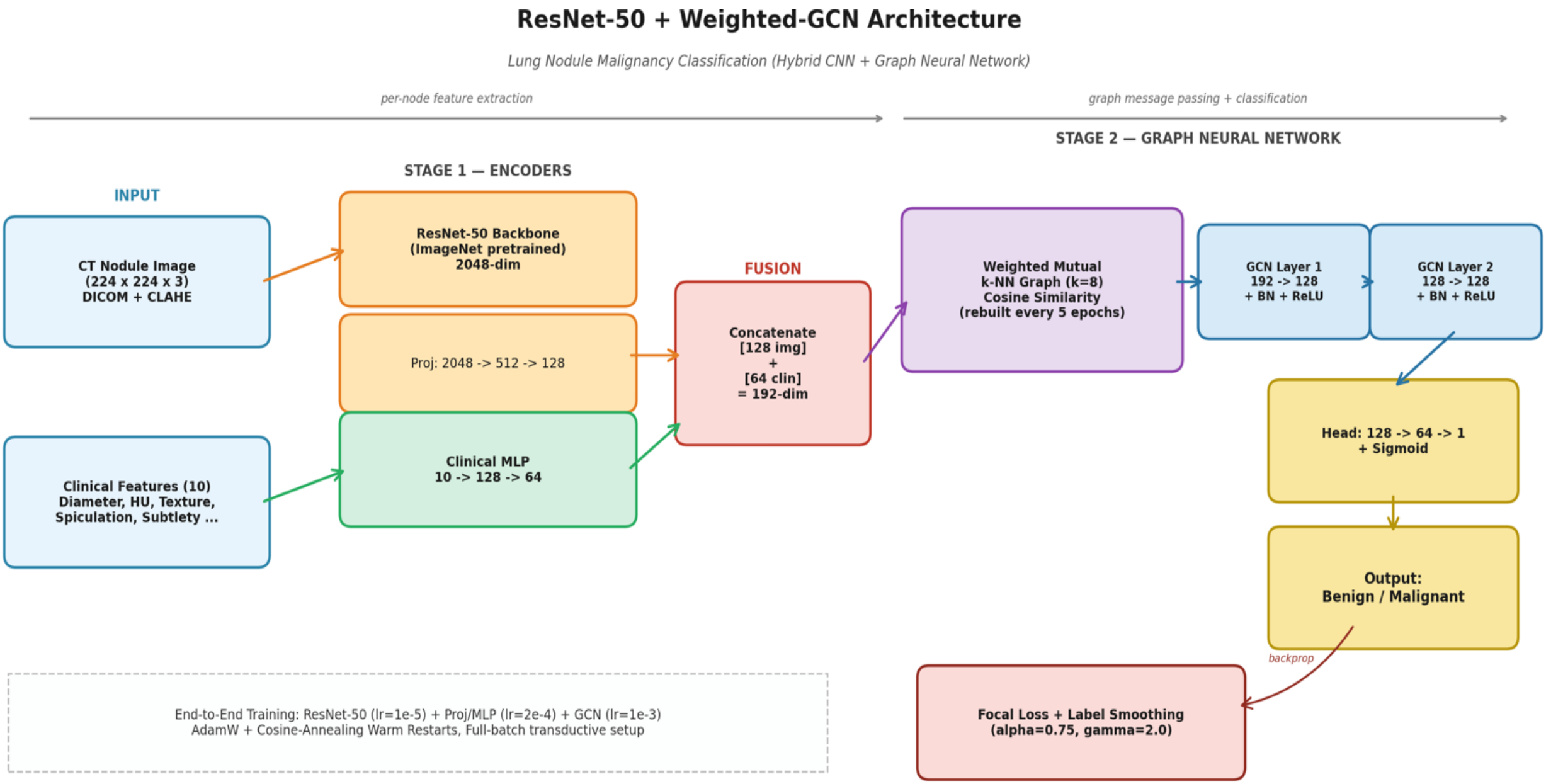
FMCIB image-only is the **best model on pathology**

| | radiologist-consensus AUC | pathology AUC | Δ (pathology – radiologist) |
|---|---|---|---|
| Med3D × image | 0.612 | 0.470 | −0.142 |
| Med3D × image + attrs | 0.960 | 0.726 | **−0.234** |
| Med3D × full | 0.954 | 0.679 | **−0.275** |
| FMCIB × image | 0.885 | **0.887** | **+0.002** |
| FMCIB × image + attrs | 0.960 | 0.729 | **−0.231** |
| FMCIB × full | 0.952 | 0.679 | **−0.274** |

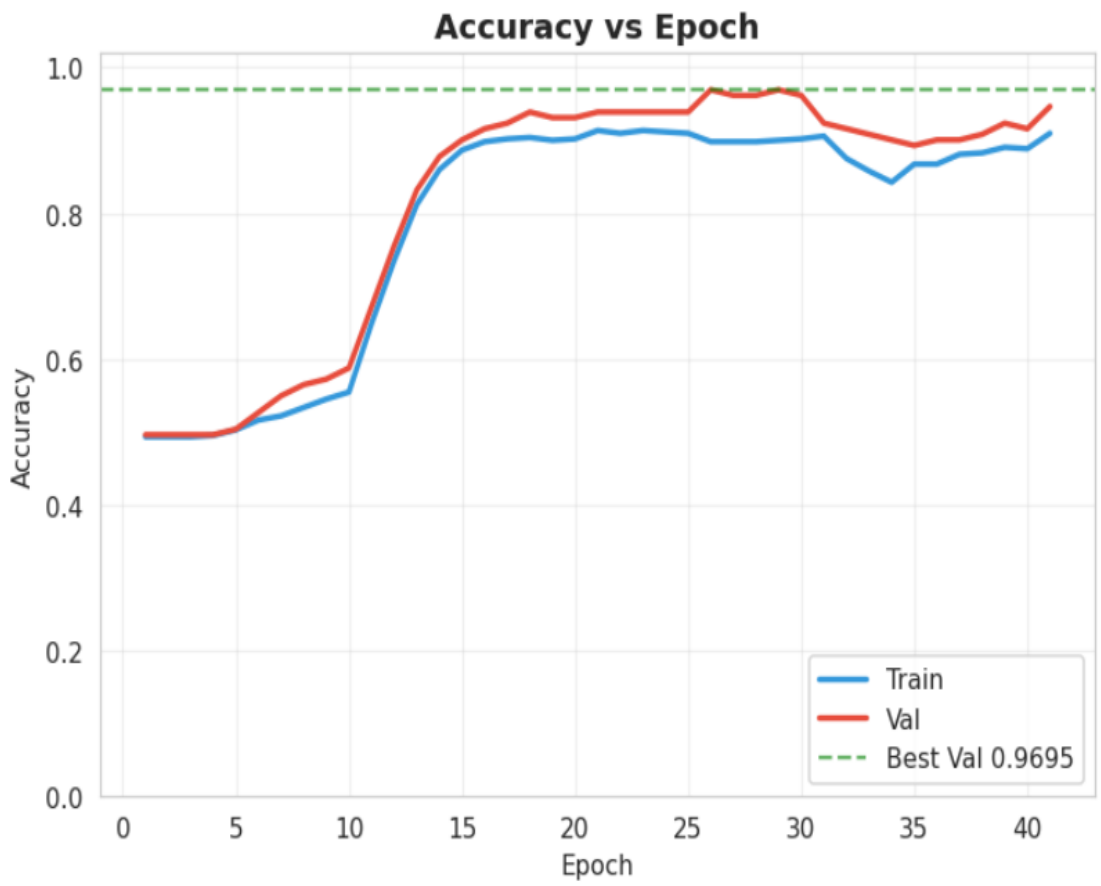
Re-running Experiment 3 against pathology ground truth

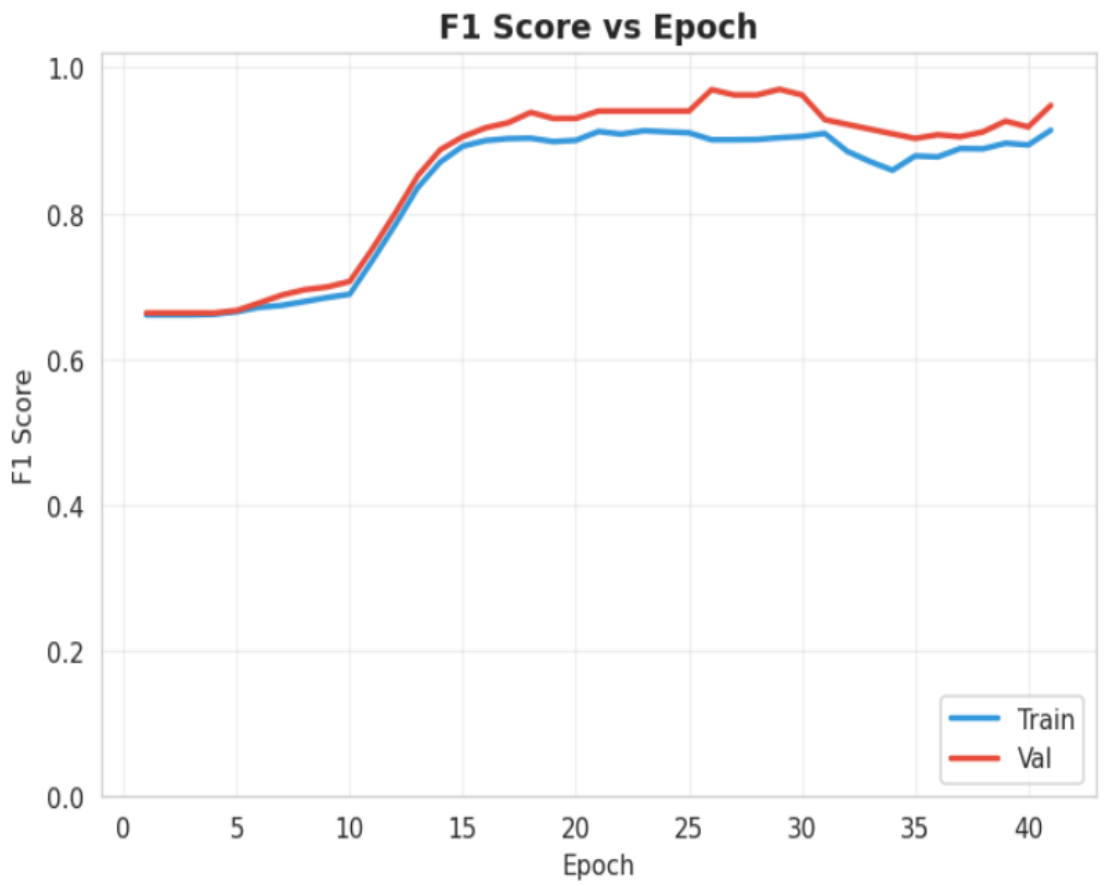
Key takeaway: FMCIB × image is the only cell that **learns something about malignancy rather than something about radiologist opinion.**

# Experiment-4 Pre work

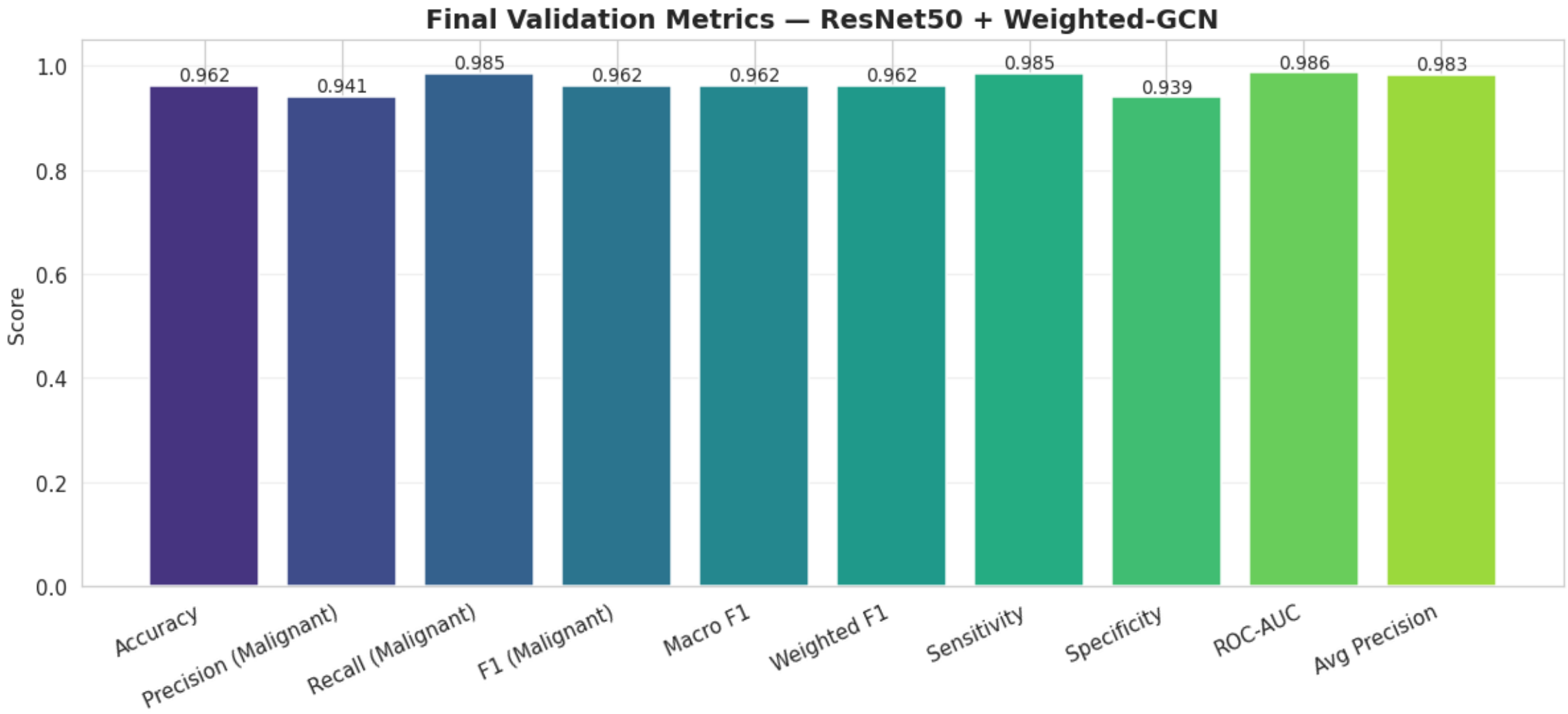


# Experiment-4 Pre work

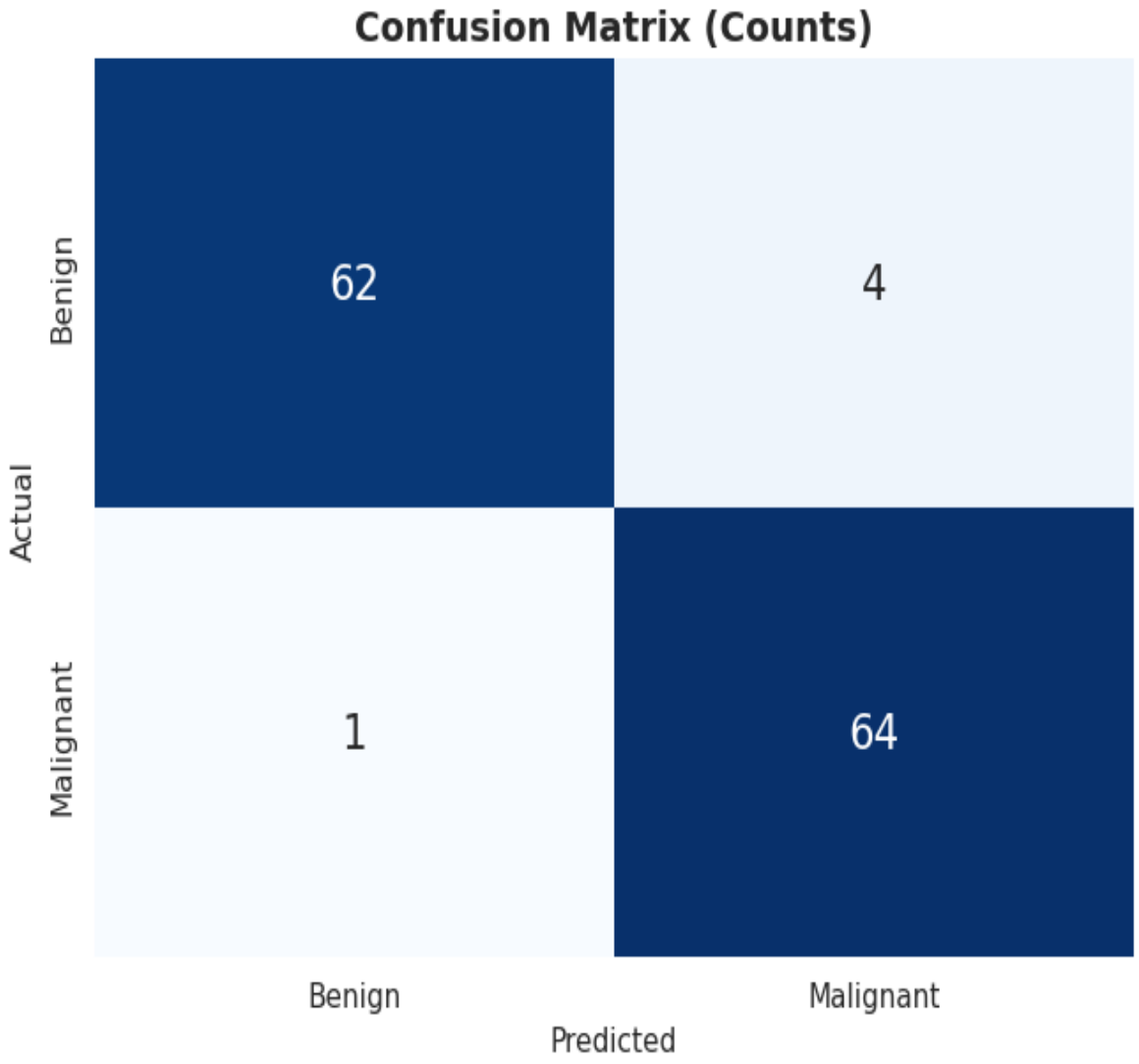


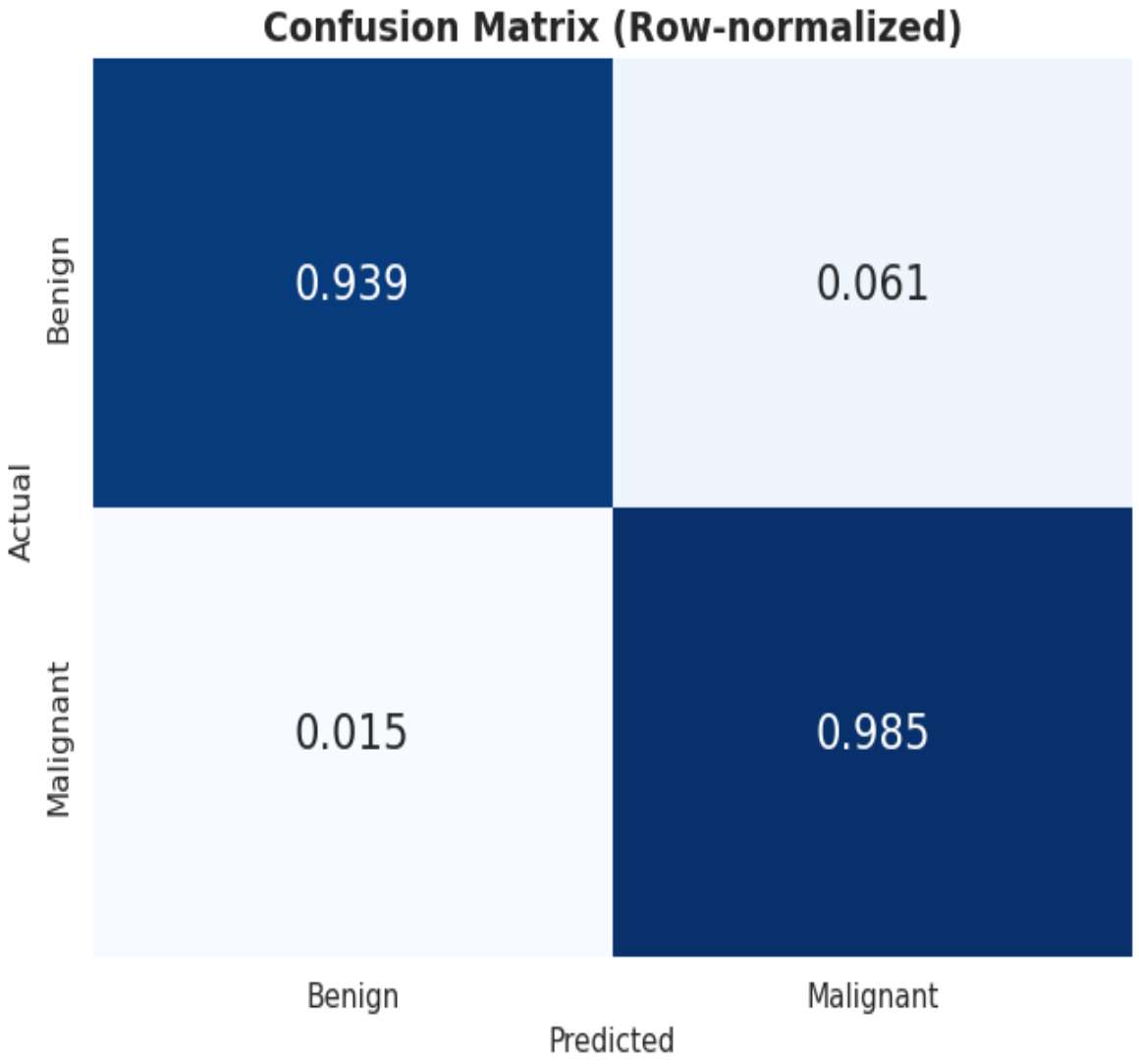



# Experiment-4 Pre work



# Experiment-4 Pre work



# Experiment 4. Graph Transformer

FUSION

**⊕ Feature Fusion (torch.cat)**

img_emb(128-d) ‖ clin_emb(64-d)

**fused node feature: 192-d (one vector per nodule)**

GRAPH CONSTRUCTION

**Mutual k-NN Graph Builder (k=8)**

kneighbors_graph(cosine distance) → symmetrise → isolated-node repair

edge_weight = 1 − cosine_dist (clamped ≥ 1e-4) | graph rebuilt every 5 epochs

**N nodes (all nodules) · edge_index: (2,E) · edge_weight: (E,) → E ≈ 8N**

GRAPH TRANSFORMER

**Graph Transformer Layer 1 — TransformerConv (Shi et al., 2021)**

in_channels=192 · heads=4 · head_dim=32 · concat=True · edge_dim=1 · beta=True

$$x_i' = W_1 x_i + \sum_{j \in \mathcal{N}(i)} \alpha_{ij}(W_2 x_j + W_6 e_{ij}) \quad \alpha_{ij} = \mathrm{softmax}\left[\frac{(W_3 x_i)^\top (W_4 x_j + W_6 e_{ij})}{\sqrt{d_{head}}}\right]$$

edge_attr = edge_weight.unsqueeze(-1) → projected via $W_6$ → added to attention score AND value

BatchNorm1d(128) → ReLU → Dropout(0.3)

**output: 128-d (32 per head × 4 heads, concatenated)**

**Graph Transformer Layer 2 — TransformerConv**

in_channels=128 · heads=4 · head_dim=32 · concat=True · edge_dim=1 · beta=True

Same attention mechanism — edge weights fully re-used

BatchNorm1d(128) → ReLU

**output: 128-d**

CLASSIFICATION HEAD

**Classification Head**

Linear(128 → 64) · ReLU · Dropout(0.3)

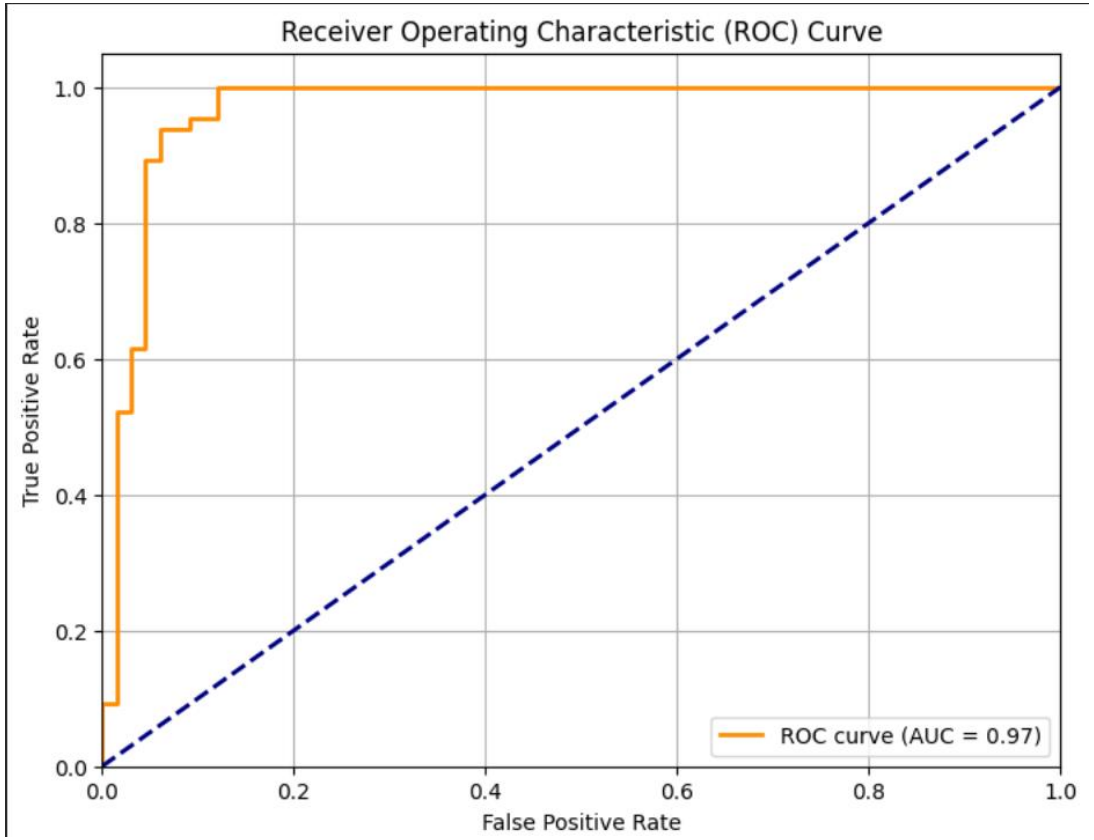
Linear(64 → 1) → σ(logit)

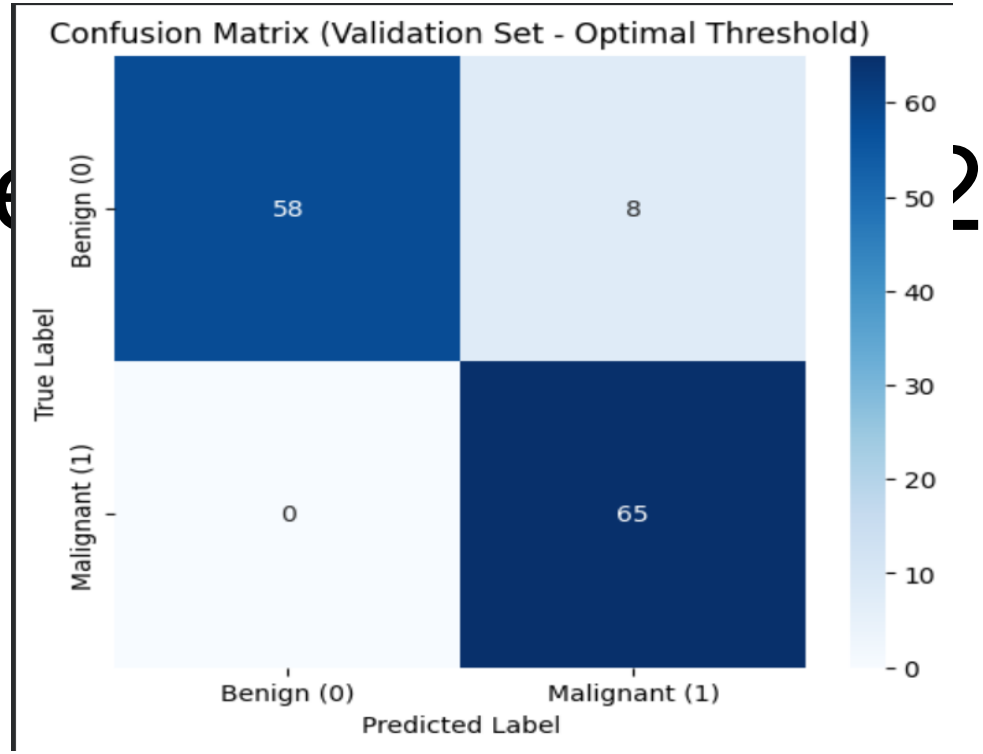
**Loss: FocalLoss (α=0.75, γ=2.0, label_smoothing=0.05)**

**TRAINING CONFIG**

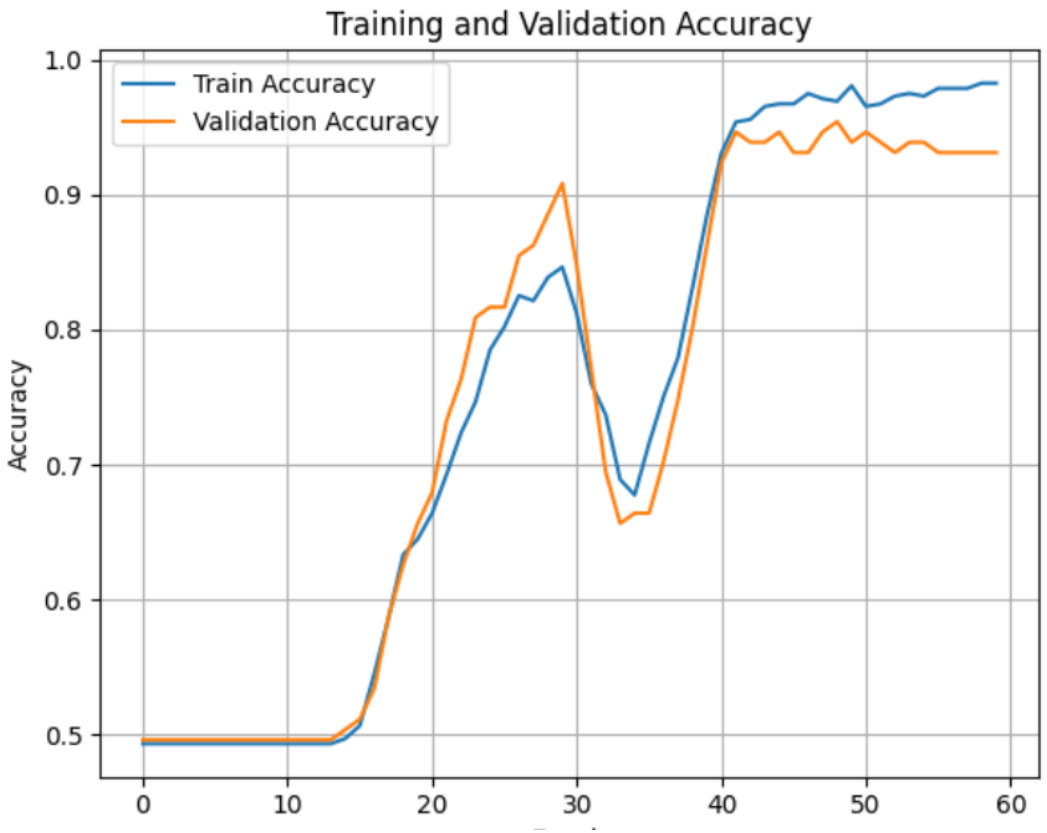
| | |
|---|---|
| Epochs | **60** |
| Patience | **15** |
| Batch size | **16** |
| Backbone LR | **1e-5** |
| Proj/Clin LR | **2e-4** |
| GNN LR | **1e-3** |
| Weight decay | **1e-4** |
| Grad clip | **2.0** |
| Scheduler | **CosineAnneal** |
| $T_0$ / T_mult | **10 / 2** |
| η_min | **1e-6** |

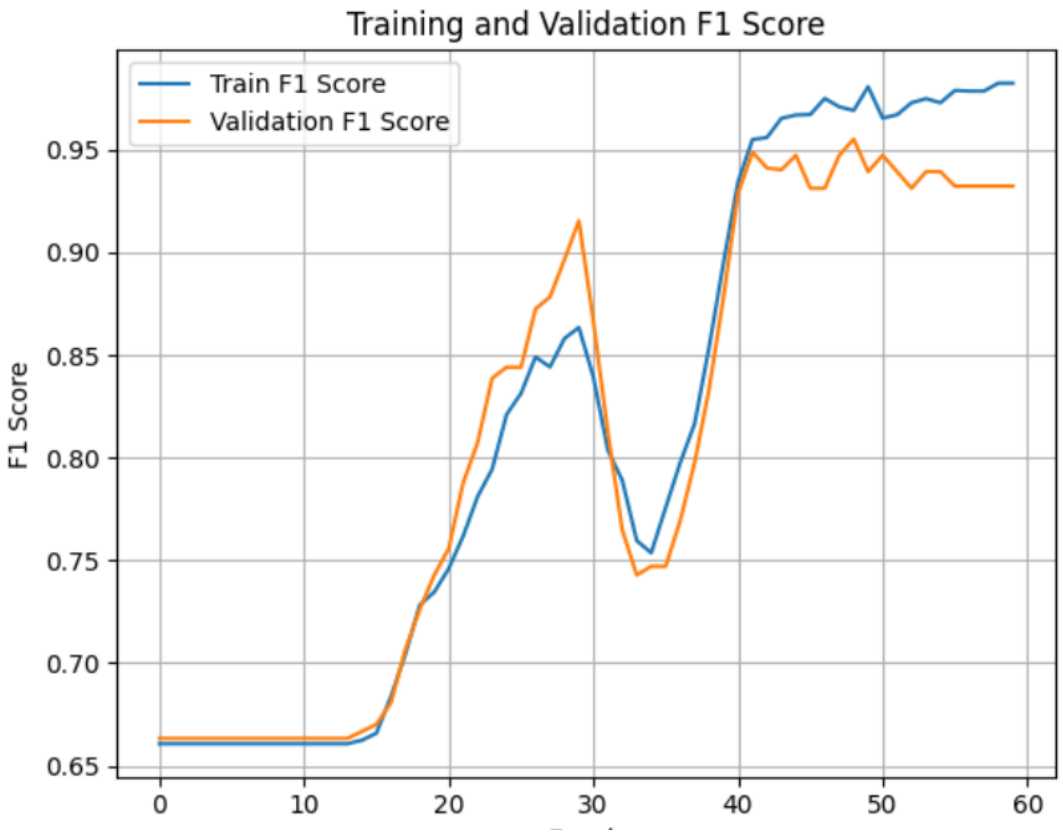
# Experiment 4, Graph Transformer Results



# Graph Transformer Training and Validation Curve



# Why switching to LUNA25 is risky

LUNA25 changes the data contract: scans are distributed as NIfTI-based challenge data with more than 4,000 CT exams and over 6,000 nodule annotations, so file layout, storage, and preprocessing assumptions from LIDC break silently rather than fail loudly.

The biggest early failure modes are mismatched volume folders, preprocessed intensity ranges instead of raw HU, and world-coordinate to voxel-centering errors, which can make patches blank or off-target and leave validation near chance even when the model code is technically running.

# Next Steps

- **Interpret and consolidate results of independent tests for final report**

- **Adopt FMCIB × image (no attributes) as the primary configuration for Experiment 4 (LUNA25 transfer).** It's the only configuration that generalizes across label sources and therefore the only one that can be expected to generalize across datasets. LUNA25 doesn't have LIDC's per-nodule attribute ratings anyway, so a model that depends on the attribute shortcut would have to be retrained from scratch.

- **Report pathology AUC in every future write-up.** Radiologist-consensus AUC is a performance proxy; pathology AUC is the thing that matters. The gap between them is the most honest measure of attribute leakage.

- **Consider re-running Experiment 2 on FMCIB × image.** With the attribute shortcut removed, there's now real residual signal for a graph to add. The "GCN underperforms MLP" verdict from Experiments 1–2 was obtained under attribute-saturated features; it hasn't been tested in the FMCIB × image regime where the image branch actually learns something.

# References


1. Kipf, T. N. & Welling, M. (2017), "Semi-Supervised Classification with Graph Convolutional Networks," *ICLR 2017*. — Foundational GCN architecture for semi-supervised node classification on graphs; defines the message-passing framework used in this project.

2. Chen, S. et al. (2019), "Med3D: Transfer Learning for 3D Medical Image Analysis," *arXiv:1904.00625*. — Provides the Med3D pretrained 3D ResNet-50 weights (trained on 23 medical segmentation datasets) used as the frozen image feature extractor.

3. Ma, X. et al. (2023), "A novel fusion algorithm for benign-malignant lung nodule classification on CT images," *BMC Pulmonary Medicine*, 23:462. — Closest existing work using GCN for lung nodule classification on LIDC-IDRI (AUC 0.9629). Uses GCN for feature fusion across CNNs; our approach differs by using GCN for node classification on a nodule-similarity graph with multi-modal features.
4. https://aim-harvard.github.io/foundation-cancer-image-biomarker/